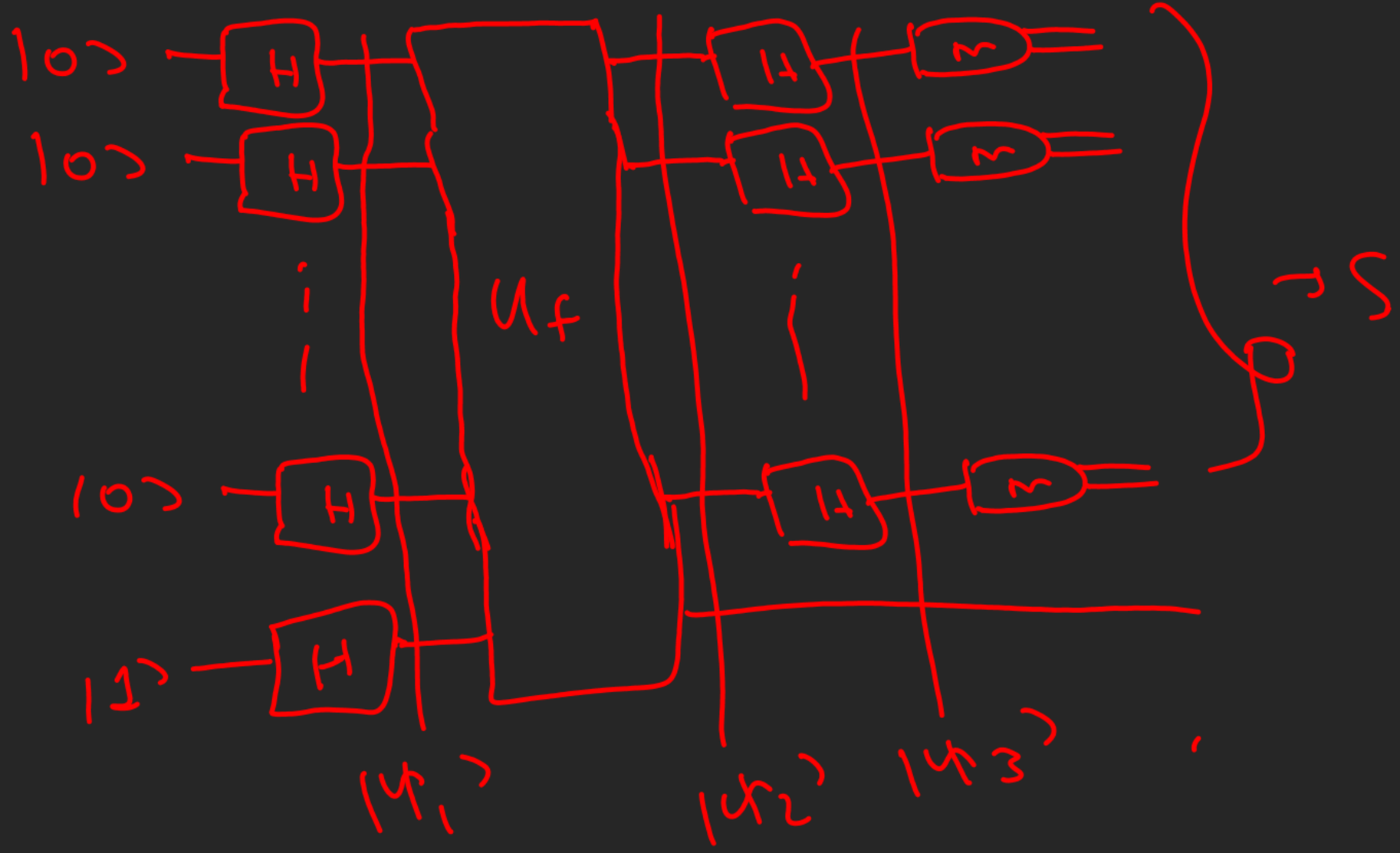
Bernstein-Vazirani Problem

Input → $f: \Sigma^n \to \Sigma$

Promise → There exists a binary string $s = s_{n-1} \cdots s_0$ for which $f(x) = s \cdot x$ for all $x \in \Sigma^n$

Output → The string s


|0⟩
|0⟩
|0⟩
|1⟩
H
H
H
H
U_f
H
H
M
M
M
|y⟩
|y₁⟩
|y₂⟩
|y₃⟩

$$|\psi_3\rangle = \frac{1}{2^n} \sum_{x \in \{0,1\}^n} \sum_{y \in \{0,1\}^n} (-1)^{f(x) + x \cdot y} |y\rangle \otimes |-\rangle$$

$$f(x) = s \cdot x \qquad (-1)^{s \cdot x \oplus x \cdot y} = (-1)^{(s \oplus y) \cdot x}$$

$$|\psi_3\rangle = \frac{1}{2^n} \sum_{x \in \{0,1\}^n} \sum_{y \in \{0,1\}^n} (-1)^{(s \oplus y) \cdot x} |y\rangle$$

$$\frac{1}{2^n} \sum_{x \in \{0,1\}^n} (-1)^{z \cdot x} = \begin{cases} 1 & \text{if } z = 0^n \\ 0 & \text{if } z \neq 0^n \end{cases}$$

$s \oplus y = 0^n$ if and only if $s = y$

$= |4_3\rangle = |s\rangle \otimes |-\rangle$

↳ measure it to get s

→ Any probabilistic algorithm at least takes n queries to get s.